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Windows PowerShell
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PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W6D3_LangChain_Tools_Agents> python w6d3_langchain_tools_agents.py

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TASK 1: LANGCHAIN CHAIN
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Input 1: Artificial Intelligence
Output: Artificial Intelligence (AI) is a type of computer technology that allows machines to think and learn like humans, enabling them to perform tasks such as image recognition, speech analysis, and decision-making. AI systems can be trained on vast amounts of data, allowing them to improve their performance over time and make predictions or decisions based on patterns they've learned.

Input 2: Machine Learning
Output: Machine Learning (ML) is a type of artificial intelligence that allows computers to learn and improve from data without being explicitly programmed. It enables computers to identify patterns, make predictions, and classify information based on the data they're trained on, similar to how humans learn and adapt. This process is automated, allowing machines to become more accurate over time with minimal human intervention.

Input 3: LangChain
Output: LangChain is an open-source framework that allows developers to build decentralized applications (dApps) on the blockchain, particularly Ethereum. It simplifies the process of integrating blockchain technology with web development, enabling creators to create scalable and user-friendly dApps with ease. By providing a set of pre-built components and tools, LangChain helps bridge the gap between traditional web development and blockchain development.

Input 4: Python
Output: Python is a programming language that allows users to create computer programs that can perform tasks automatically. It's easy to learn and use, making it a popular choice for beginners and experts alike. Python is often used for tasks like data analysis, artificial intelligence, and web development, but its versatility makes it suitable for many other applications as well.

Input 5: Cloud Computing
Output: Cloud computing is a way of storing and accessing data online, rather than on your own device or computer. You pay for what you use, rather than owning the entire system, which makes it more cost-effective and flexible. Think of it like renting a room instead of buying a house - you only pay for what you need, when you need it.

TASK 1 STATUS: COMPLETED
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TASK 1 STATUS: COMPLETED
Chain tested successfully with 5 inputs.

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TASK 2: CONVERSATION MEMORY / HISTORY
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Turn 1
User: My name is Sanjana.
Assistant: Hello Sanjana! Unfortunately, this is our first conversation, so we don't have any previous discussion to refer to. How can I assist you today?

Turn 2
User: What is my name?
Assistant: Hello Sanjana! According to our previous conversation, you told me that your name is indeed Sanjana.

Turn 3
User: I am learning LangChain.
Assistant: Hello again Sanjana! According to our previous conversation, I mentioned that according to our previous conversation, you told me that your name is indeed Sanjana.

Turn 4
User: What am I learning?
Assistant: You're learning LangChain.

Turn 5
User: What do you know about me from this conversation?
Assistant: From our conversation, I know that:

1. Your name is Sanjana.
2. You are learning LangChain.

These are the only two pieces of information I have gathered so far about you.

TASK 2 STATUS: COMPLETED
Conversation history maintained across 5 turns.
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Turn 5
User: What do you know about me from this conversation?
Assistant: From our conversation, I know that:

1. Your name is Sanjana.
2. You are learning LangChain.

These are the only two pieces of information I have gathered so far about you.

TASK 2 STATUS: COMPLETED
Conversation history maintained across 5 turns.

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TASK 3: LANGCHAIN AGENT
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Agent Task 1
User: Calculate 25 * 16.
Agent: The result of 25 * 16 is 400.

Agent Task 2
User: Search the web for information about LangChain.
Agent: LangChain is an open-source, Python-based framework for building custom blockchain explorers, nodes, and other tools. It allows developers to create applications that interact with blockchains in a more efficient and scalable way.

Key Features:

1. Modular Architecture: LangChain is designed with modularity in mind, allowing developers to easily add or remove features as needed.
2. Customizable: Users can customize the framework to fit their specific use case, whether it's building a blockchain explorer or creating a node for a particular blockchain.
3. Scalability: LangChain aims to provide high-performance and scalability, making it suitable for large-scale applications.
4. Blockchain Support: Currently supports several blockchains, including Ethereum, Solana, Polkadot, and more.

Applications:
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Applications:

1. Blockchain Explorers: Users can create custom explorers that allow users to browse and interact with blockchain data in a more user-friendly way.
2. Node Development: LangChain enables developers to build nodes for specific blockchains, which are essential components of the overall blockchain ecosystem.
3. Smart Contract Development: Developers can leverage LangChain's capabilities to create smart contracts that run on various blockchains.

LangChain is an innovative framework that provides users with a powerful toolset for building custom blockchain applications and tools.

Agent Task 3
User: Calculate (100 + 50) / 5.
Agent: The result of the calculation is 30.0.

TASK 3 STATUS: COMPLETED
Agent tested successfully with 3 tasks.
Agent contains 2 tools: Web Search Stub + Calculator.

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W6D3 PRACTICAL TASK SUMMARY
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[COMPLETED] Task 1
PromptTemplate -> Ollama LLM -> OutputParser
Tested with 5 inputs

[COMPLETED] Task 2
Conversation history / memory
Verified across 5 conversation turns

[COMPLETED] Task 3
LangChain Agent
Two tools:
1. Web Search Stub
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Agent Task 3
User: Calculate (100 + 50) / 5.
Agent: The result of the calculation is 30.0.

TASK 3 STATUS: COMPLETED
Agent tested successfully with 3 tasks.
Agent contains 2 tools: Web Search Stub + Calculator.

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W6D3 PRACTICAL TASK SUMMARY
=====

[COMPLETED] Task 1
PromptTemplate -> Ollama LLM -> OutputParser
Tested with 5 inputs

[COMPLETED] Task 2sss
Conversation history / memory
Verified across 5 conversation turns

[COMPLETED] Task 3
LangChain Agent
Two tools:
  1. Web Search Stub
  2. Calculator
Tested with 3 tasks

W6D3 PRACTICAL TASKS: COMPLETED

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END OF W6D3 PRACTICAL EXECUTION
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PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W6D3_LangChain_Tools_Agents>
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